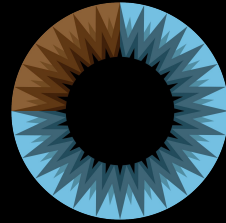


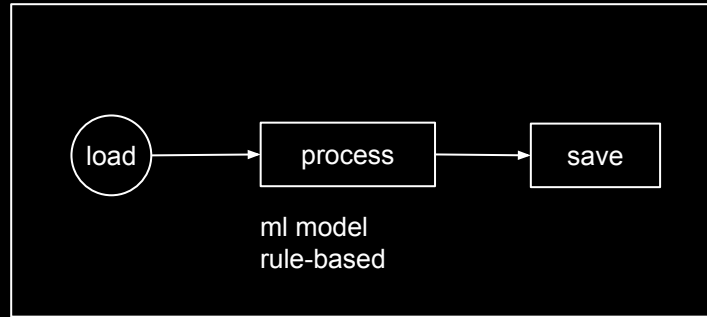
MACHINE LEARNING

Highly recommended for
background information



3Blue1Brown's YouTube Course on Neural
Networks and Deep Learning

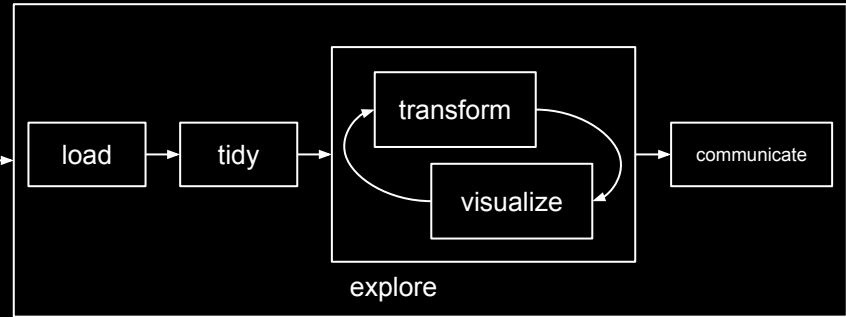
pre-process
unstructured data



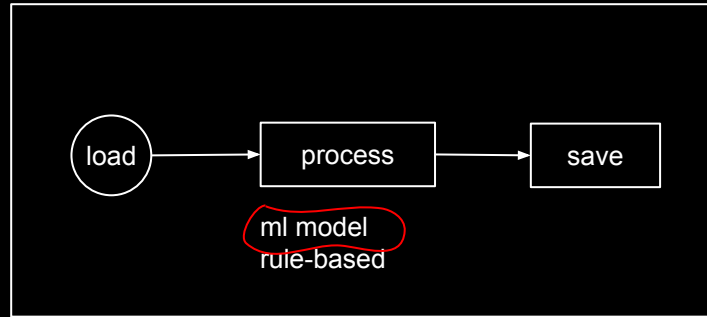
program



exploratory data
analysis



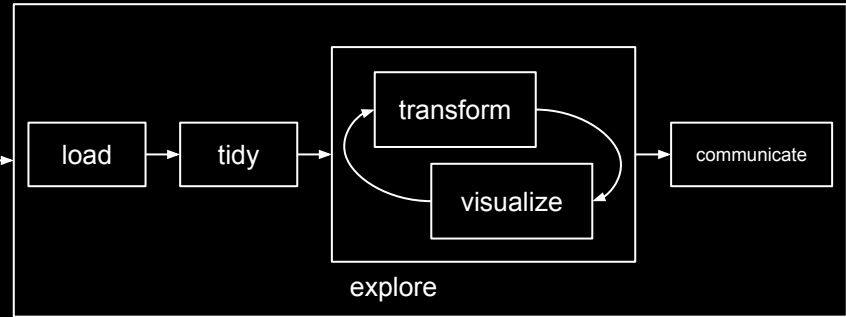
pre-process
unstructured data



program



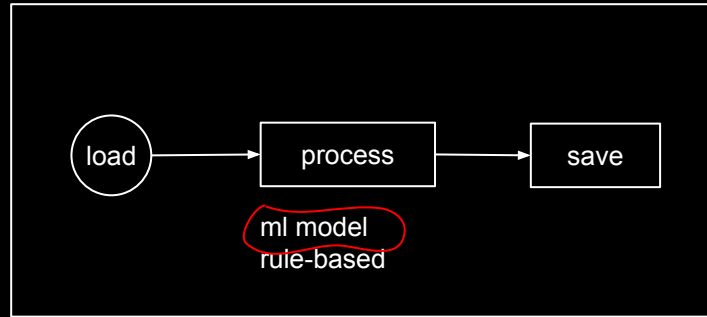
exploratory data
analysis



program



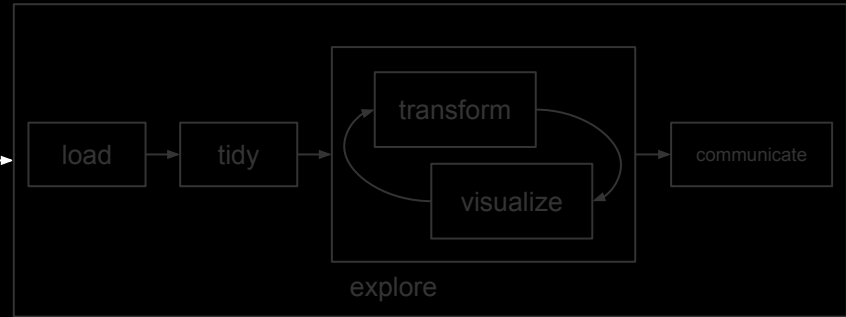
pre-process
unstructured data



program

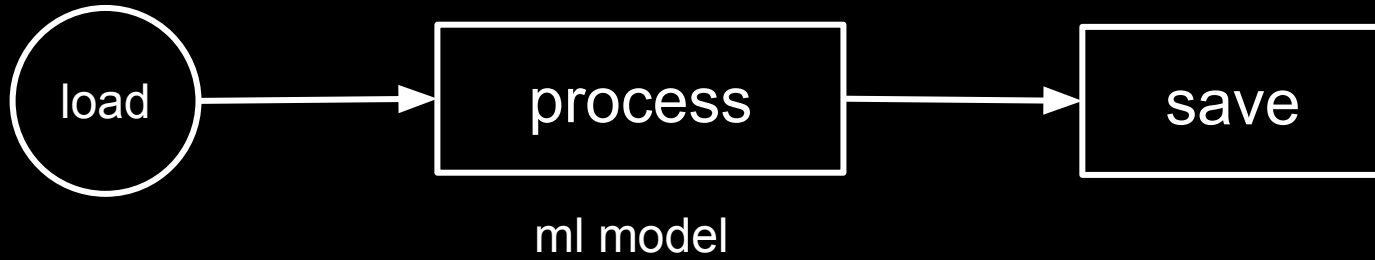


exploratory data
analysis



program





program



YouTube



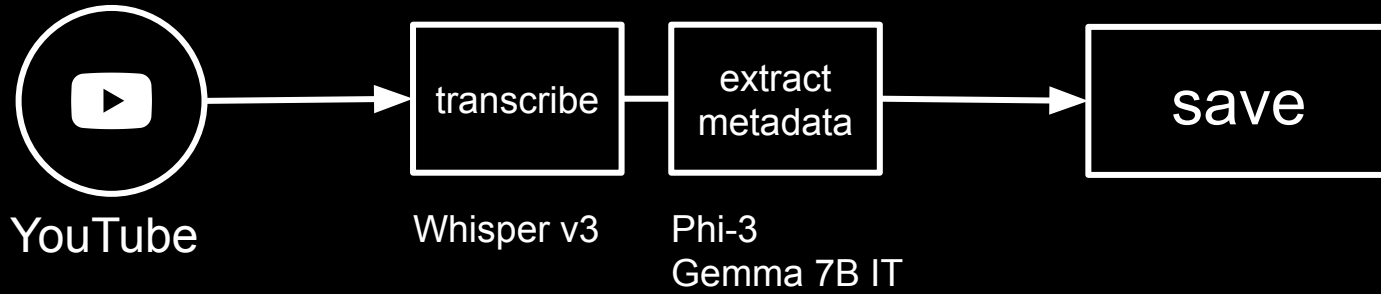
process

ml model

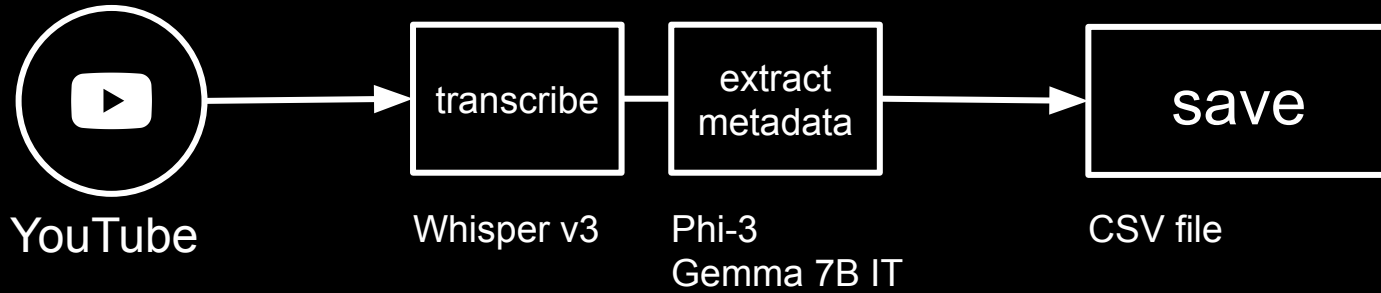


save

program



program



program

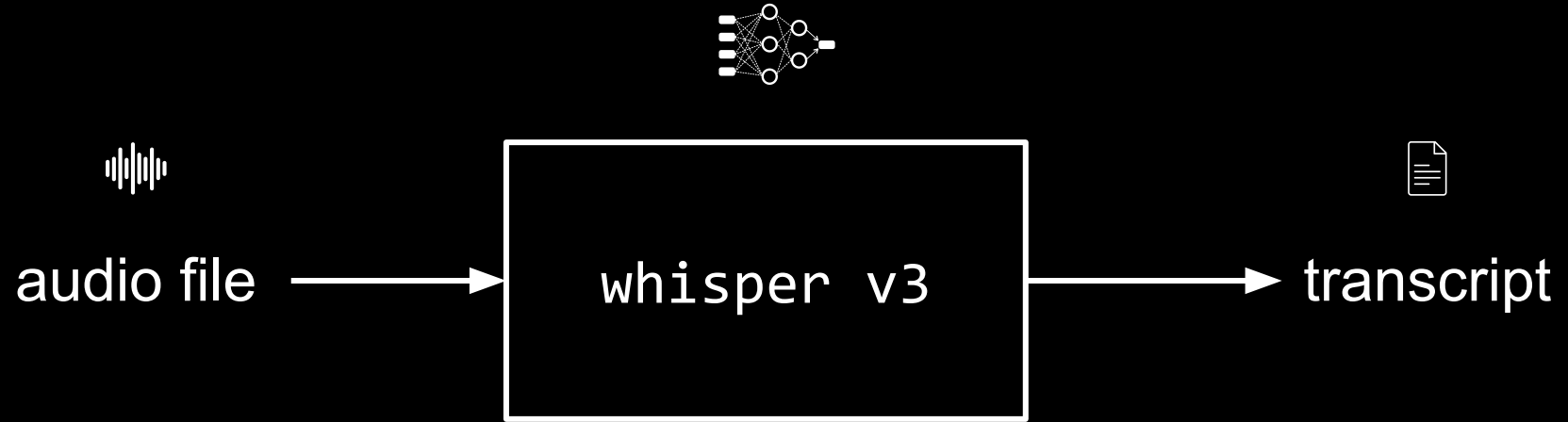
YouTube API

Whisper v3

<https://arxiv.org/abs/2212.04356>



<https://huggingface.co/openai/whisper-large-v3>



Large Language Models (LLMs)

what has been said so far?
(*prompt / context*)

what has been said so far?
(*prompt / context*)



prediction of next token based on
learnt probability distribution

what has been said so far?
(*prompt / context*)



prediction of next token based on
learnt probability distribution

+

(randomness)

what has been said so far?
(*prompt / context*)



prediction of next token based on
learnt probability distribution

+

(randomness)

+

(filter)

(*discriminating, insulting content*)

what has been said so far?
(*prompt / context*)



prediction of next token based on
learnt probability distribution

+

(randomness)

+

(filter)

(*discriminating, insulting content*)



next word (*token*)

what has been said so far?
(*prompt / context*)



prediction of next token based on
learnt probability distribution

+

(randomness)

+

(filter)

(*discriminating, insulting content*)



next word (*token*)



PROMPTING



basic elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

basic elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

Explain the binary number system.

basic elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

Explain the binary number system.

start simple

basic elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

You are a friendly tutor and your task is to explain complex concepts as simple as possible.

Explain the binary number system.

basic elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

You are a friendly tutor and your task is to explain complex concepts as simple as possible.

Your answers are never longer than 10 sentences.

Explain the binary number system.

ZERO-SHOT PROMPTING

basic elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

Classify the text into neutral,
negative or positive.

Text: "What a great dinner!"

Sentiment:

basic elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

Classify the text into neutral,
negative or positive.

Text: "What a great dinner!"

Sentiment:

this will be replaced with
data later...

FEW-SHOT PROMPTING

IN-CONTEXT LEARNING

examples in the context to learn from

Extract all references to countries and their continent in the following text using the format from the examples below.

Example 1: "They played the team called 'Die Mannschaft' in the world cup final"

Correct answer: Germany, Europe

Example 2: "The Three Lions once again lost to Germany in a semi final"

Correct answer: England, Europe, Germany, Europe

Text: "The Selecao was destroyed 1:7 by the DFB selection in their home stadium."

Answer:

examples in the context to learn from

Extract all references to countries and their continent in the following text using the format from the examples below.

Example 1: "They played the team called 'Die Mannschaft' in the world cup final"

Correct answer: Germany, Europe

Example 2: "The Three Lions once again lost to Germany in a semi final"

Correct answer: England, Europe, Germany, Europe

Text: "The Selecao was destroyed 1:7 by the DFB selection in their home stadium."

Answer:

more prompting strategies

chain-of-thought (CoT)

self-consistency

generate knowledge prompting

prompt chaining (subtasks)

tree-of-thoughts (ToT)

retrieval-augmented-generation (RAG)

...

Phi-3

<https://arxiv.org/abs/2404.14219>



~~<https://huggingface.co/microsoft/Phi-3-mini-128k-instruct>~~

<https://huggingface.co/microsoft/Phi-3-medium-128k-instruct>

Gemma 2B / 7B Instruct

<https://arxiv.org/abs/2403.08295>



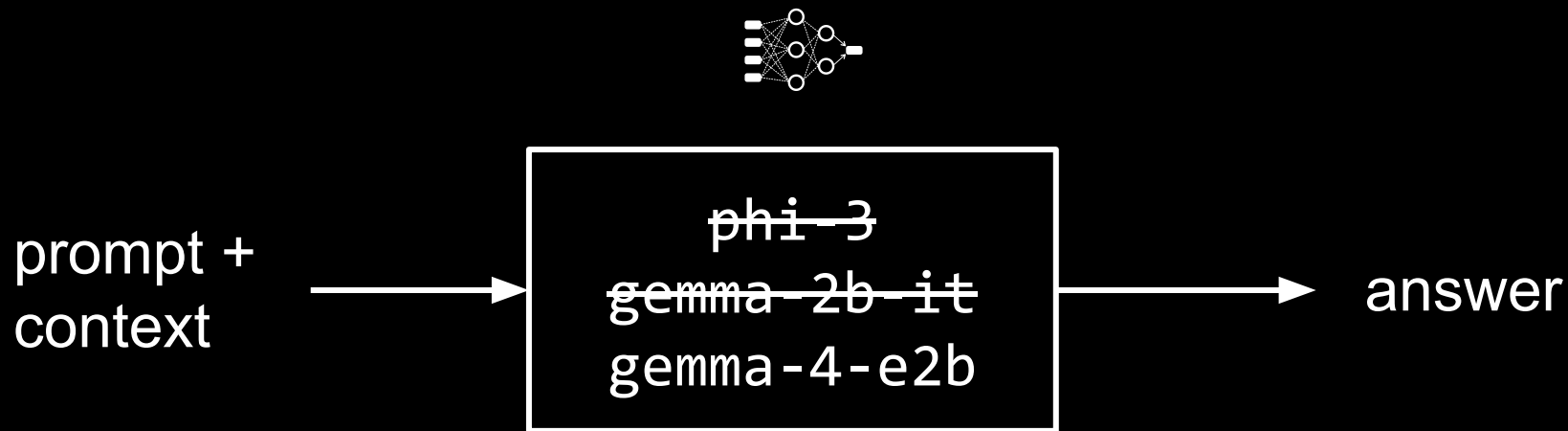
<https://huggingface.co/google/gemma-2b-it>

<https://huggingface.co/google/gemma-7b-it>

Gemma 4 E2B



<https://huggingface.co/google/gemma-4-E2B>

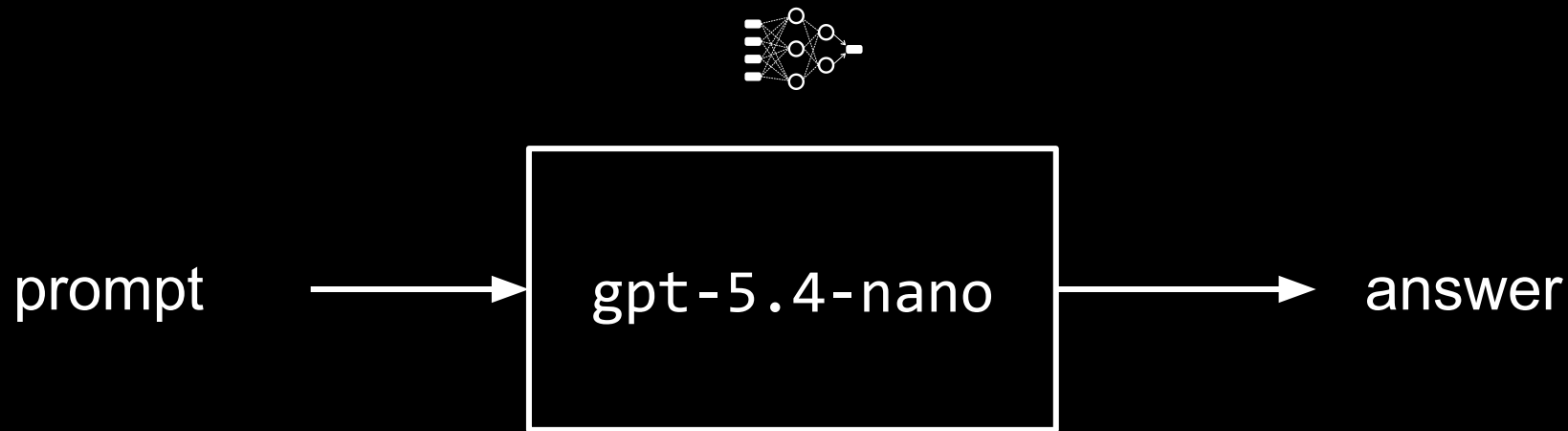


OpenAI's GPT-5.4 family

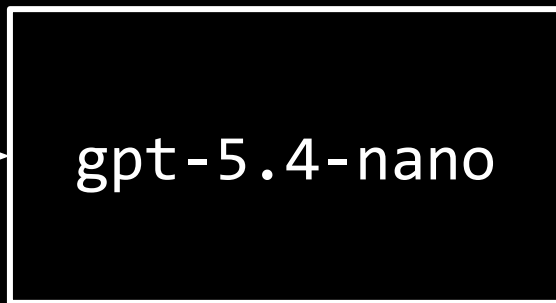
<https://platform.openai.com/docs/models>

gpt-5.5		gpt-5.4-mini		gpt-5-nano	
A new class of intelligence for coding and professional work.		Our strongest mini model yet for coding, computer use, and subagents		Fastest, most cost-efficient version of GPT-5	
Learn more		Learn more		Learn more	
Playground		Playground		Playground	
Reasoning	● ● ● ● ●	Reasoning	● ● ● ● ●	Reasoning	● ●
Speed	⚡ ⚡ ⚡ ⚡	Speed	⚡ ⚡ ⚡ ⚡	Speed	⚡ ⚡ ⚡ ⚡ ⚡
Input	📄 📄 📄 📄	Input	📄 📄 📄 📄	Input	📄 📄 📄 📄
Output	📄 📄 📄 📄	Output	📄 📄 📄 📄	Output	📄 📄 📄 📄
Reasoning tokens	✔	Reasoning tokens	✔	Reasoning tokens	✔
PRICING	PER 1M TOKENS	PRICING	PER 1M TOKENS	PRICING	PER 1M TOKENS
Input	\$5.00	Input	\$0.75	Input	\$0.05
Cached Input	\$0.50	Cached Input	\$0.08	Cached Input	\$0.01
Output	\$30.00	Output	\$4.50	Output	\$0.40
CONTEXT		CONTEXT		CONTEXT	
Window	1,050,000	Window	400,000	Window	400,000
Max Output Tokens	128,000	Max Output Tokens	128,000	Max Output Tokens	128,000
Knowledge Cutoff	Dec 01, 2025	Knowledge Cutoff	Aug 31, 2025	Knowledge Cutoff	May 31, 2024

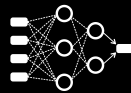
<https://developers.openai.com/api/docs/models>



prompt +
image



answer



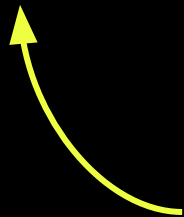
prompt +
image



gpt-5.4-nano



answer



multimodality

local open-weight vs. cloud models